American Computer Science League

2020-2021 ● Contest 2: Solutions ● Classroom Division

1. Prefix-Infix-Postfix

3 6 * 2 / 3 9 + 4 / − 2 3 ^ +
= (3 * 6) 2 / (3 + 9) 4 / − (2 ^ 3) +
= (18 / 2) (12 / 4) − 8 =
= (9 − 3) 8 +
= 6 + 8 = 14

1. 14 (D)

2. Prefix-Infix-Postfix

$$\frac{A^2 + B}{C} + ABC$$

= (A^2 + B) / C + A*B*C
= (A 2 ^) B +) / C + ((A B *) C *)
= (A 2 ^ B + C /) + A B * C *
= A 2 ^ B + C / A B * C * +

2. A2^B+C/AB*C*+ (B)

3. Prefix-Infix-Postfix

− / * 4 + 2 3 * 2 5 / + ^ 6 2 4 8
= − / * 4 (+ 2 3) (* 2 5) / + (^ 6 2) 4 8
= − / (* 4 5) 10 / (+ 36 4) 8
= − (/ 20 10) (/ 40 8)
= − 2 5 = -3

3. -3 (D)

4. Prefix-Infix-Postfix

$A = 2\pi rh + 2\pi r^2$:
A = 2 * P * r * h + 2 * P * r^2
A = (* 2 P) * r * h + (* 2 P) * (^ r 2)
A = (* (* 2 P *) r) * h + (* (* 2 P) (^ r 2))
A = (* (* (* 2 P) r) h) + (* (* 2 P) (^ r 2))
A = (+ * * * 2 P r h * * 2 P ^ r 2)
The complete formula is: = A + * * * 2 P r h * * 2 P ^ r 2

4. =A+***2Prh**2P^r2 (C)

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5. Bit-String Flicking

(01011 OR (NOT 10010) AND 00011)
= 01011 OR (01101 AND 00011) because AND has higher precedence
= 01011 OR 00001
= 01011

5. 01011 (A)

6. **Bit-String Flicking**

(NOT (RSHIFT-2 (LCIRC-2 10011)))
= (NOT (RSHIFT-2 01110))
= (NOT 00011)
= 11100

6. 11100 (E)

7. Bit-String Flicking

((LSHIFT-1 11011) OR (RCIRC-2 01101) AND 01111)
= (10110 OR (01011 AND 01111))
= (10110 OR 01011) = 11111

7. 11111 (A)

8. Bit-String Flicking

Let X = abcde
LHS = ((LCIRC-3 X) AND 10110)
= ((LCIRC-3 abcde) AND 10110)
= (deabc AND 10110) = d0ab0
LHS = RHS
⇒ d0ab0 = 10100
⇒ d = 1, a = 1, b = 0, c = *, e = *
⇒ 10*1*

8. 4 (C)

9. LISP

(ADD (EXP 2 3) (MULT -3 5 (SUB 5 9)) (DIV 18 3))
= (ADD 8 (MULT -3 5 -4) 6)
= (ADD 8 60 6)
= 74

9. 74 (C)

10. LISP

(CDR (CAR '((b c d) (e f) g)))
= (CDR '(b c d))
= (c d)

10. (c d) (D)